

PROJECT DOCUMENTATION: PLANT CO PERFORMANCE DASHBOARD

1. Executive Summary

The **Plant Co Performance Dashboard** is designed to provide high-level visibility into the company's financial and operational health. By centralizing Sales, Quantity, and Gross Profit metrics, this tool enables management to identify growth opportunities, pinpoint underperforming regions, and monitor the efficiency of the product lifecycle.

2. Data Engineering & Transformation

Before building the visuals, the raw data underwent a rigorous cleaning and modeling process to ensure a "Single Source of Truth."

Data Cleaning (Power Query)

- **Duplicate Removal:** Ensured every transaction in the Fact_sales table is unique to prevent inflated revenue figures.
- **Null Handling:** Fixed missing values in key columns to maintain calculation integrity.
- **Data Typing:** Standardized currencies and date formats for seamless cross-table filtering.

Custom Calendar Development

To support advanced Time Intelligence (YTD and PYTD), a standardized Gregorian calendar was created.

- **Range:** January 1, 2022, to December 31, 2024.

- **Purpose:** This allows the model to compare any date this year with the equivalent date last year, even during leap years or partial months.
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3. Analytical Logic (DAX Architecture)

The "intelligence" of the dashboard is driven by custom DAX measures designed for accuracy and user flexibility.

The Synchronized Comparison (Inpast)

- **Formula:** `Dim_Date[Date] <= LatestSalesdatePY`
- **The "Why":** This creates a "cut-off" point. It ensures that when we look at "Last Year," we only look at the days that have already passed in the "Current Year," preventing an unfair comparison between a full year and a partial year.

Core Financial Measures

- **Sales:** `SUM(Fact_sales[Sales_USD])` — Total revenue generated.
- **COGS (Cost Price):** `SUM(Fact_sales[cogs])` — The direct cost to purchase/produce the plants sold.
- **Gross Profit:** `[Sales] - [COGS]` — The profit remaining after covering the cost price, used to pay for overhead and rent.

Dynamic Switching Engine

- **Mechanism:** Created a Switch Table to allow users to toggle the entire dashboard between Sales, Quantity, and Gross Profit.
- **Benefit:** This provides a 3-in-1 reporting experience, keeping the interface clean while offering deep drill-down capabilities.

4. Visualization & Business Insights

Chart Component	Technical Purpose	Business Conclusion
KPI Scorecards	Compares S_YTD vs S_PYTD.	Gives an instant "Red or Green" status on whether the company is beating last year's performance.
Waterfall Chart	Tracks variance by Country and Month.	Identifies exactly <i>which</i> country (e.g., Germany or Colombia) is responsible for a dip in profit.
Scatter Plot	Maps Account Profitability (GP% vs Volume).	Distinguishes between high-value clients and those who are buying high volumes at dangerously low margins.
TreeMap (Top/Bottom 10)	Rank-based filtering of regions.	Forces management to focus on the 10 most critical areas that need immediate operational improvement.

5. Strategic Conclusions

- 1. Standardized Comparison:** By using the Inpast filter, the dashboard provides a mathematically honest Year-over-Year (YoY) growth figure.
- 2. Profitability Focus:** The shift from tracking "Sales" to "Gross Profit" ensures the business prioritizes making money over just moving inventory.

3. **Data-Driven Decisions:** The dynamic "Switch" allows different departments (Sales, Finance, Operations) to use the same dashboard for their specific needs.
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